Towards Constructing A URI Synonym Dictionary

Make datasets more easily Findable, Accessible, Interoperable, and Reusable

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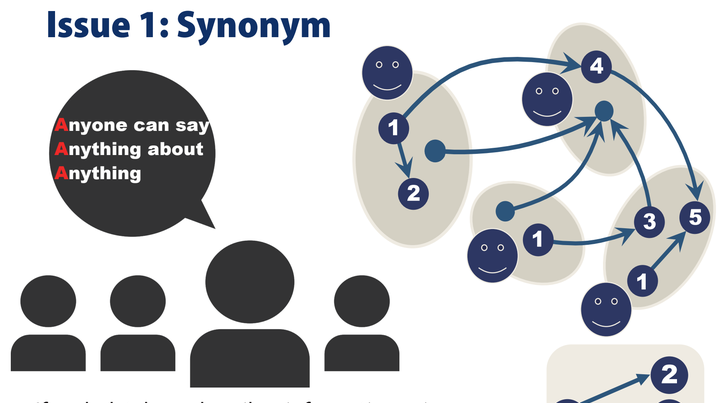
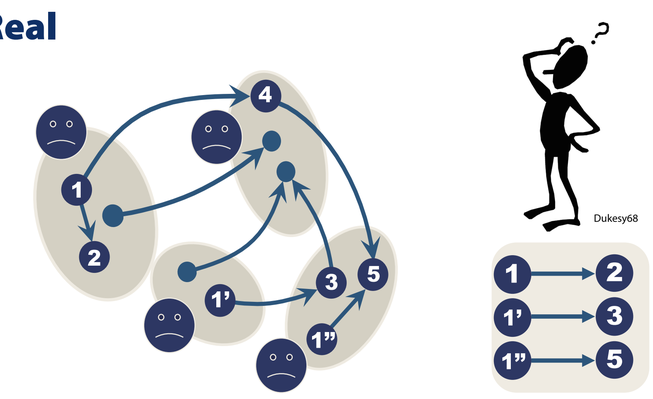
Database Center for Life Science (DBCLS) provides RDF Portal (https://rdfportal.org/) that currently hosts 61 RDF datasets in the life sciences domain including ontologies, with the total number of triples exceeding 124 billion. Among them, there are numerous cases where a same concept is represented by different URIs, which hinders data integration through URIs, one of the advantages of the RDF model. To address this issue of synonymous URIs, initiatives such as Identifiers.org, TogoID, and Biomappings have been undertaken. Although these attempts provide useful resources, yet another type of synonymous URI database is needed for RDF Portal to realize semantically integrated access. Therefore, efforts are underway to build a synonymous URI database within RDF Portal, leveraging these existing data resources. We believe our efforts contribute to realizing junctions connecting various life science knowledge graphs.

Issue 1: Synonym



If each database describes information using a common identifier (URI) for a concept, it enables efficient cross-database information integration.

In Real



If different identifiers are used for the same concept, concepts that should be connected may be treated as distinct, potentially leading to the loss of valuable information. In the example above, nodes such as 1 and 3 or 1′ and 2 should be connected, but they remain unlinked due to different identifiers. In this case, nodes1, 1′ , and 1″ are synonymous with each other.

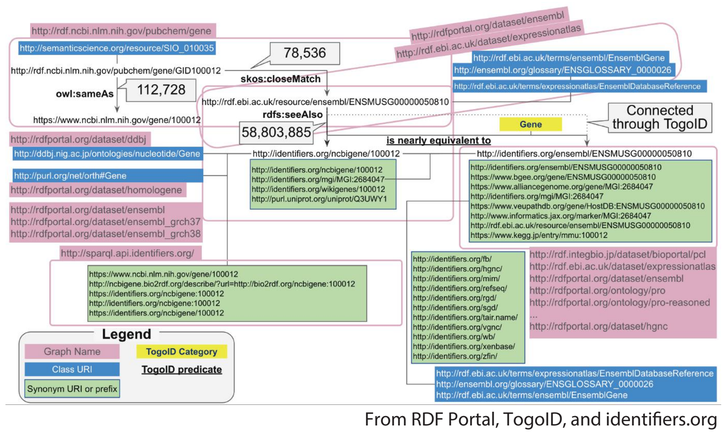
Examples

Below we found in the RDF Portal (https://rdfportal.org/)

- **http://identifiers.org/ensembl/ENSG00000136997**
- **http://icgc.link/Gene/ENSG00000136997**
- **http://rdf.ebi.ac.uk/resource/ensembl/ENSG00000136997**
- **http://bio2rdf.org/ensembl:ENSG00000136997**
- **http://rdf.ebi.ac.uk/resource/ensembl/ArrayExpress/ENSG00000136997**
- **http://purl.uniprot.org/ensembl/ENSG00000136997**
- **http://omabrowser.org/ontology/oma#GENE_ENSG00000136997**
- **https://bgee.org/bgee15_0/gene/ENSG00000136997**

Issue 2: Structure

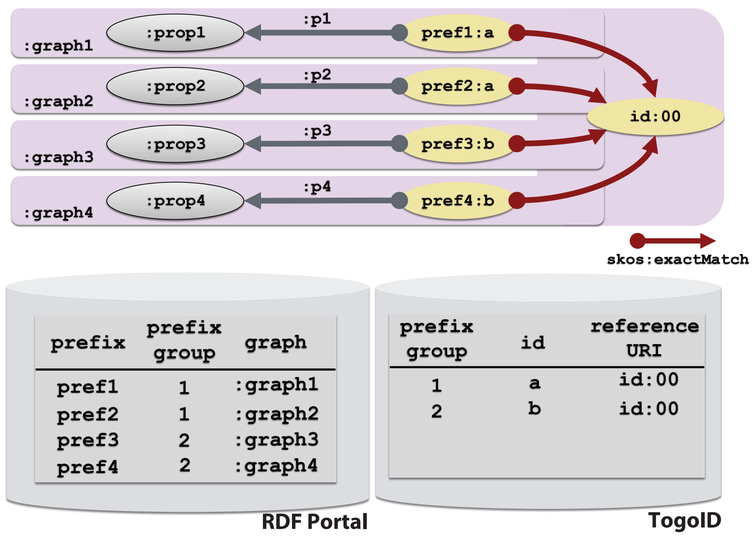
Some key URIs are referred as objects by several subject URIs. This leads to an issue that we cannot efficiently get the properties comprehensively.



From RDF Portal, TogoID, and identifiers.org

We want to "describe ncbigene:100012" and retrieve related properties all at once.

Our Plan

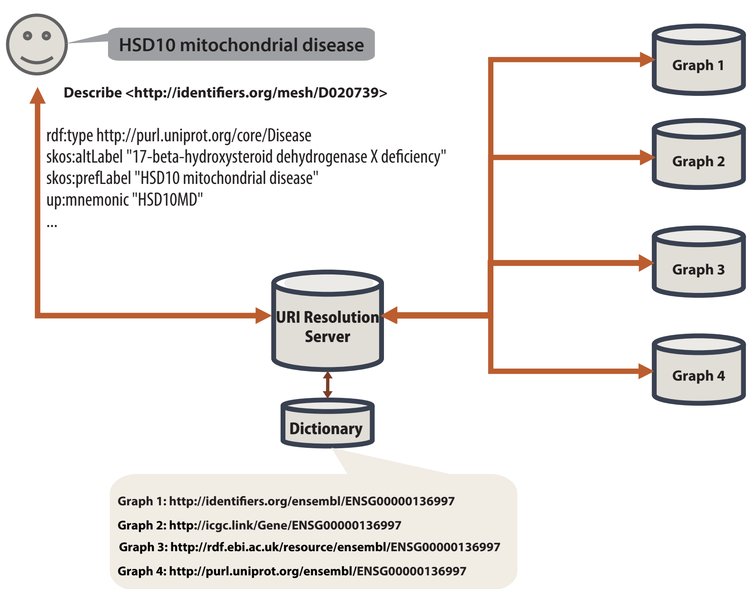


In this example, all the URIs from **pref1:a** to **pref4:b** are synonym to each other. The URIs **pref1:a** and **pref2:a** share their local ID portions. Similary, **pref3:b** and **pref4:b** share, too. However, there are no relationships in notation between **pref1:a** and **pref3:b**. From TogoID, we can see there are synonymous relationships between them.

So, we make dictinoaries consisting of two tables. One is to map a prefix to its prefix group and graph name stored in RDF Portal. The other is to map a prefix group and each ID to its reference URI, stored in TogoID.

The overall value of interconnected databases increases as more datasets reference related data each other.
In the case of RDF, reusing URIs.
↓
Using the same URI for the same concept.

Our Proposing System



A synonym dictionary and a URI resolution server enable to retrieve properties comprehensively.

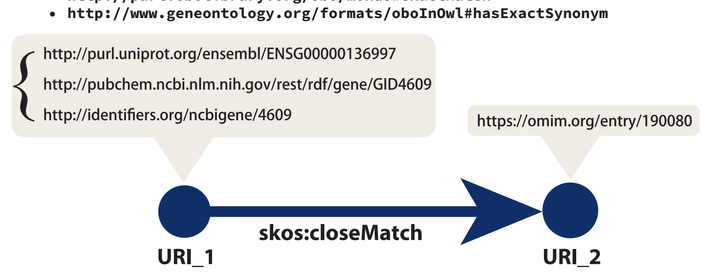
For each synonym URI group, we assign a canonical URI, and the dictionary holds key-value pairs where keys are canonical URIs and values are synonym URIs. Taking the above plan as an example, **id:00** is a canonical URI, and **:a** and **:b** are its synonyms.

The resolution server takes an URI and looks up if it is a canonical one or a synonym to a canonical one. If neither of these, it is used as is. If it is canonical or synonymous, the server obtains all the properties of the canonical URI.

Collecting Synonym URIs

Retrieve all triples connected by the predicates listed below from all RDF data stored in RDF Portal and construct a dictionary for each graph.

- `http://www.w3.org/2000/01/rdf-schema#seeAlso` (rdfs:seeAlso)
- `http://www.w3.org/2004/02/skos/core#exactMatch` (skos:exactMatch)
- `http://www.w3.org/2004/02/skos/core#closeMatch` (skos:closeMatch)
- `http://www.w3.org/2002/07/owl#sameAs` (owl:sameAs)
- `http://purl.obolibrary.org/obo/mondo#exactMatch`
- `http://www.geneontology.org/formats/oboInOwl#hasExactSynonym`



As a principle, canonical URIs belong to either identifiers.org, purl.uniprot.org, or pubchem.ncbi.nlm.nih.gov/rest/rdf .

Current Status

The numbers of retrieved triples are shown in the table below (grouped by RDF Portal triple store, sorted in ascending order of frequency, as of the end of August 2024). The names at the first column represent the main datasets stored in each triple store. The content will be reviewed to determine the presence of synonymous relationships and to establish links to the namespaces mentioned above, such as identifiers.org.

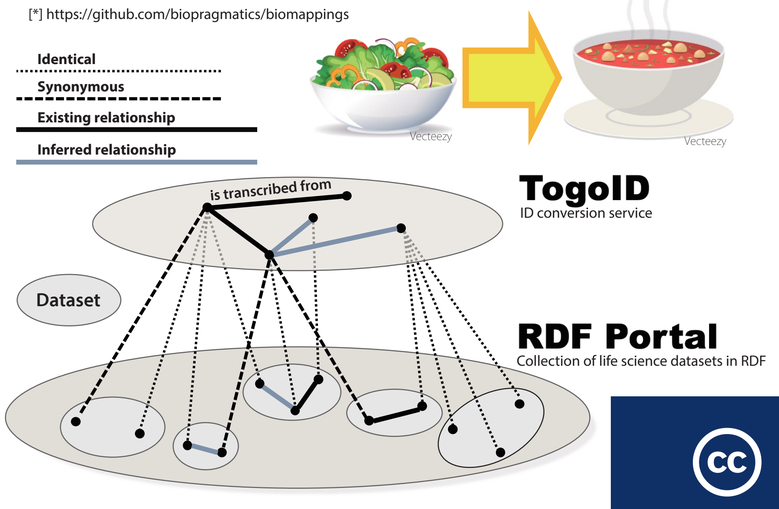
pubchem	5,918,318
pdb	63,410,584
ncbi	125,005,528
ebi	167,695,465
primary	179,221,126
kero	600,406,435
ddbj	1,319,657,122
sib	3,092,157,419
Total	5,553,471,997

This is a subset of the entire RDF Portal, focusing only on candidate triples for synonymous relationships so that the subset can be stored in a single triple store, enabling efficient searches. This data will be used for investigating synonymous URI relationships and building dictionaries. Properties for each URI can be retrieved using federated queries.

Future Works

After verifying the actual synonymous relationships from the retrieved data, we will explore the possibility of periodical dictionary update, such as workflow automation. For example, if the entity is a gene ID, we may retrieve the class URI of the synonymous URI candidate and check whether the class URI or label contains the word "gene." Additionally, external resources such as identifiers.org and Biomappings* will be referenced to assess validity. We will also investigate whether there are triples using predicates other than the six mentioned above that indicate synonymous relationships. Furthermore, as illustrated in the figure below, we will leverage the TogoID service (https://togoid.dbcls.jp/), which catalogs relationships between various life science identifiers, to enhance the RDF Portal's data with more biologically meaningful connections.

[*] https://github.com/biopragmatics/biomappings



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